

AURIX™ TC39xBD

Safety Package Release Note

About this document

Scope and purpose

The product(s) listed in this safety package release note has been developed and is released as ISO 26262:2018 Safety Element out of Context for safety requirements up to ASIL D.

If the product is intended to be used in a functional safety application, you are obliged to ensure conformity with the relevant requirements of ISO 26262:2018.

In particular you are obliged to obtain the available product safety documents and to respect the information and information updates provided by Infineon Technologies.

In the event that you merely resell the product as a distributor, you are obliged to ensure that your customers obtain the available product safety documents and that your customers respect the information and information updates provided by Infineon Technologies.

Information updates are provided via Product Change Notification (PCN), Info Note or Early Problem Notification (EPN) or any other written communication to customers or distributors that have ordered the Product within the last 5 years. The complete updated safety document is available via the Infineon sales office.

Intended audience

This safety package release note is written for experienced hardware, software and functional safety engineers involved in the design or development of a safety-related system according to ISO 26262 which consider to integrate the product into their (sub-) system. It is the responsibility of the integrator to ensure that the product is suitable for the chosen application and complies with the applicable application standards.

This document is intended for customers who have already integrated or would be integrating (in future) the AURIX™ TC39xBD in their safety application.

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Introduction

1 Introduction

Please read this document carefully and be sure you understand the information provided before integrating the AURIX™ TC39xBD in a safety-related application.

1.1 Document overview

This document summarizes the set of customer relevant documentation as referenced in the customer safety case document and the corresponding valid version.

An overview regarding the dedicated content is provided in chapter [2].

For dedicated document content change please always refer to the revision history of the respective document.

This document summarizes the the list of applicable product variants and derivatives.

Safety Package Release Content

2 Safety Package Release Content

The following tables provide an overview of all the documentation contained in the Safety Package Release including the corresponding valid version.

2.1 List of customer documentation

Table 1 Documents and revision numbers

Document Number	Document	Revision Number & Release Date
1	AURIX™ TC3xx IFX User Manual	V2.0 / 02-2021
2	AURIX™ TC39x Appendix to User's Manual	V2.0 / 02-2021
3	AURIX™ TC3xx ED Target Specification ¹	V3.2 / 08-2020
4	TC39xBD_BC_DS	V1.2 / 03-2021
5	TC39x_Data_Sheet_Addendum	V1.6 / 11-2020
6	AURIX™ TC3xx Safety Manual	V2.0 / 03-05-2021
7	TC39xBD_Errata_Sheet	V1.8 / 25-03-2022
8	AURIX_TC39x_FMEDA	V1.03_ext / 15-12-2020
9	CPU SBST User Manual	V1.0.3.1.0 / 03-05-2019
10	SPU SBST User Manual	V1.0.2.1.0 / 09-03-2019
11	A2G_AURIX Safety Analysis Summary Report	V1.17 / 07-05-2021
12	TC39xBD_SafetyCaseAddendum	V1.1 / 05-2021
13	TC39xBD_SafetyPackageReleaseNote	V1.2 / 04-2022
14	TriCore TC1.6.2 Core Architecture Manual; Core Architecture (Vol. 1)	V1.2.2 / 15-01-2020
15	TriCore TC1.6.2 Core Architecture Manual; Instruction Set (Vol. 2)	V1.2.2 / 15-01-2020

¹Distribution under NDA, only relevant for tool development not for application development

2.2 Applicable product variants

This Safety Package Release Note applies to the AURIX™ TC39xBD variants given in below table:

Table 2 AURIX™ TC39xBD product variants and derivatives

Name	Description and features	Package	Marking
TC39x BD	-40 °C to 150 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-516	SAL-TC399XX-256F300S BD
	-40 °C to 150 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-516	SAL-TC399XP-256F300S BD

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-40 °C to 150 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-292	SAL-TC397XP-256F300S BD
-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-516	SAK-TC399XP-256F300S BD
-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-516	SAK-TC399XX-256F300S BD
-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-292	SAK-TC397XP-256F300S BD
-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-292	SAK-TC397XA-256F300S BD
-40 °C to 125 °C; Quad Core Emulation Device; 10 MB Flash; 300 MHz	PG-LFBGA-292	SAK-TC397QA-160F300S BD
-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-292	SAK-TC397XX-256F300S BD
-40 °C to 125 °C; Quad Core Emulation Device; 12 MB Flash; 300 MHz	PG-LFBGA-292	SAK-TC397QP-192F300S BD
-40 °C to 125 °C; Quad Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-292	SAK-TC397QP-256F300S BD
-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-292	SAK-TC397XZ-256F300S BD
-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-292	SAK-TC397XM-256F300S BD
-40 °C to 150 °C; Quad Core Emulation Device; 12 MB Flash; 300 MHz	PG-LFBGA-292	SAL-TC397QP-192F300S BD
-40 °C to 150 °C; Quad Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-292	SAL-TC397QP-256F300S BD
-40 °C to 150 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-292	SAL-TC397XZ-256F300S BD
-40 °C to 150 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-292	SAL-TC397XX-256F300S BD

List of changes compared to previous version

3 List of changes compared to previous version

3.1 Changes in customer documents

Table 3 List of document changes compared to previous version

Document	Safety related	Change summary
AURIX™ TC3xx IFX User Manual	N/A	Initial Version
AURIX™ TC39x Appendix to User's Manual	N/A	Initial Version
AURIX™ TC3xx ED Target Specification ¹	N/A	Initial Version
TC39xBD_BC_DS	N/A	Initial Version
TC39x_Data_Sheet_Addendum	N/A	Initial Version
AURIX™ TC3xx Safety Manual	N/A	Initial Version
TC39xBD_Errata_Sheet	Yes	New document version "v1.8" is available, for more details please refer to the document Revision History
AURIX_TC39x_FMEDA	N/A	Initial Version
CPU SBST User Manual	N/A	Initial Version
SPU SBST User Manual	N/A	Initial Version
A2G_AURIX Safety Analysis Summary Report	N/A	Initial Version
TC39xBD_SafetyCaseAddendum	N/A	Initial Version
TC39xBD_SafetyPackageReleaseNote	N/A	This document
TriCore TC1.6.2 Core Architecture Manual; Core Architecture (Vol. 1)	N/A	Initial Version
TriCore TC1.6.2 Core Architecture Manual; Instruction Set (Vol. 2)	N/A	Initial Version

3.2 Changes in product variants

Table 4 AURIX™ TC39xBD product variants and derivatives

Name	Description and features	Package	Marking

Revision history**Revision history**

Document version	Date of release	Description of changes
1.0	05-2021	Initial Version
1.1	12-2021	New Erratasheet document released
1.2	04-2022	New Erratasheet document released

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